

Dynamic Perfusion Representation and Aggregation Network for Nodule Segmentation Using Contrast-Enhanced US

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Abstract—Dynamic contrast-enhanced ultrasound (CEUS) imaging has been widely applied in lesion detection and characterization, due to its offered real-time observation of microvascular perfusion. Accurate lesion segmentation is of great importance to the quantitative and qualitative perfusion analysis. In this paper, we propose a novel dynamic perfusion representation and aggregation network (DpRAN) for the automatic segmentation of lesions using dynamic CEUS imaging. The core challenge of this work lies in enhancement dynamics modeling of various perfusion areas. Specifically, we divide enhancement features into the two scales: short-range enhancement patterns and long-range evolution tendency. To effectively represent real-time enhancement characteristics and aggregate them in a global view, we introduce the perfusion excitation (PE) gate and cross-attention temporal aggregation (CTA) module, respectively. Different from the common temporal fusion methods, we also introduce an uncertainty estimation strategy to assist the model to locate the critical enhancement point first, in which a relatively distinguished enhancement pattern is displayed. The segmentation performance of our DpRAN method is validated on our collected CEUS datasets of thyroid nodules. We obtain the mean dice coefficient (DSC) and intersection of union (IoU) of 0.794 and 0.676, respectively. Superior performance demonstrates its efficacy to capture distinguished enhancement characteristics for lesion recognition.

Index Terms—Dynamic contrast-enhanced US, enhancement dynamics representation, cross-view fusion.

I. INTRODUCTION

ULTRASOUND (US), as the first-line diagnostic imaging, has become increasingly important in clinical disease diagnosis and therapeutic evaluation, due to inherent advantages of cost-effectiveness, portability, non-ionizing radiation, and real-time assessment [1]. Through an intravenous injection of gas-filled microbubble contrast agent, contrast-enhanced ultrasound (CEUS) allows for a real-time assessment of blood flow through tumor microvascular system by amplifying acoustic responses coming from small vessels, as shown in Fig. 1. In the past decades, CEUS has gained wide acceptance for the lesion detection and characterization, such as focal liver lesions [2], thyroid nodules [3], breast cancer [4], etc.

For routine examination, radiologists are usually required to observe the complete perfusion process back and forth, and then contour the lesion region manually for the continuous assessment of dynamic enhancement. Fig. 1 basically covers the initial enhancement to the eventual wash out. However, manual outline of lesions requires a high level of expertise and is susceptible to subjective errors. To reduce inter-/intra- observer variability, various automatic or semi-automatic segmentation methods have been proposed for lesion boundary recognition in CEUS imaging, including active contours [5], fuzzy clustering [6], and fully convolution network (FCN) [7]. A common practice for these methods is to pre-select a reference point at which lesion has a relatively distinguished boundary. Nonetheless, unitary enhancement pattern inevitably lost dynamic enhancement characteristics for the segmentation task. On the one hand, tumor interior enhancement tends to be heterogeneous due to irregular vessel branching, high tortuosity and unpredictable vessel diameter [8], [9]. On the other hand, lesion boundary lacks a distinct transitional band as a result of tumor invasive growth to periphery [10]. Taking peripheral nodular enhancement as an example, typical enhancement characteristics *not only* consist in initial peripheral focal enhancement, *but also* in the continuous progression in a centripetal direction to partial fill-in (Fig. 1). Therefore, it is important to reason over the whole perfusion process, from initial enhancement to the eventual wash out, to distinguish lesion from normal tissues.

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Code is available <https://github.com/wanpeng16/DpRAN>.
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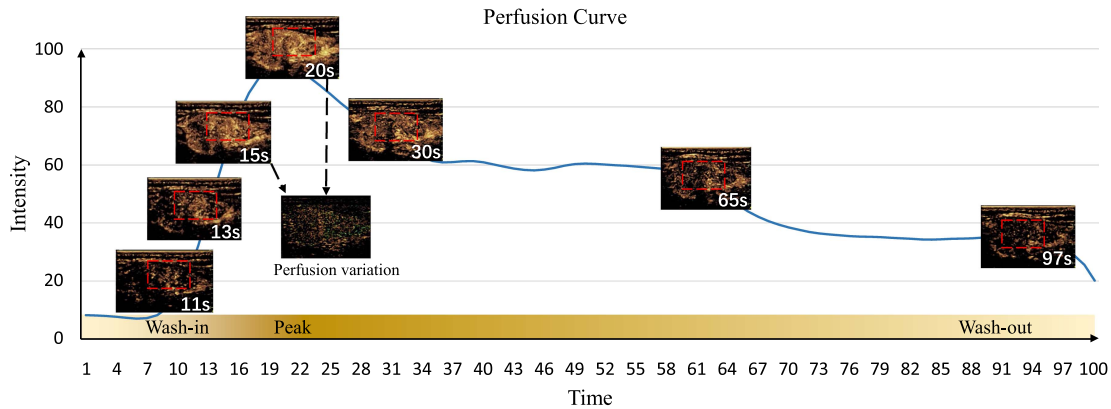


Fig. 1. An illustration of the dynamic enhancement process of thyroid nodule. A complete perfusion process can be divided into three phases ranging from the initial wash-in, reaching peak, to the eventual wash out. We plot the perfusion curve (i.e., the variation of enhancement degree over time) of the outlined regions (denoted in magenta). We could see that both enhancement pattern at the single point and contrast variations among adjacent frames are valuable to characterize perfusion discrepancy between lesion and surrounding tissues.

The primary challenge of dynamic CEUS-based segmentation is how to model dynamic enhancement evolution patterns associated with different blood supply status. With reference to the field of video understanding [11], we divide temporal features into the two scales: 1) short-range enhancement patterns among adjacent points; and 2) long-range evolution patterns over the complete examination. For simplicity, we could observe the enhancement evolution from the single factor of enhancement degree. The short-range temporal features refer to enhancement patterns within the short time interval, and the long-range features describe the global perfusion tendency (depicted in Fig. 1). Further, short-range enhancement patterns are modelled by enhancement morphology and dynamics features jointly, among which, dynamics features are specified to perfusion intensity variations among adjacent contrast frames. The motivation behind is that the hemodynamic changes within tumor and healthy tissues might result in distinguished perfusion variations.

Based on the abovementioned discussions, we propose a novel dynamic perfusion representation and aggregation network (DpRAN) for automatic lesion segmentation using contrast-enhanced US imaging. A schematic diagram of our DpRAN method is presented in Fig. 2, which decomposes the segmentation task into the two stages. With the identified critical point t , the specifically designed perfusion fusion module is used to aggregate the dual-view dynamic perfusion representations along the up-sampling path. The model performance was evaluated on the datasets of thyroid nodules. Extensive experimental results have validated the superiority of our method. Compared with the existing methods, we summarize the main advantages of our method as the three aspects.

- Different from previous static segmentation methods that necessitate a manually selected key frame, we adopt a confidence estimation strategy to identify the critical point in an online manner at Stage I. One point is assumed to encompass clearly enough enhancement boundary in relative to the rest.
- Towards a sufficient mine of local enhancement characteristics, we introduce a perfusion excitation (PE) gate which learns the enhancement variations among adjacent frames

in addition to static enhancement pattern at each single point. Compared with existing element-wise difference comparison, PE gate performs local representation similarity measure within a neighborhood.

- To capture the global enhancement evolution pattern, we design a cross-attention temporal aggregation (CTA) module to fuse the dual-view enhancement representation sequences. The main difference from existing self-attention mechanism is that this module contextualizes temporal supports centering the identified critical point instead of the dense associations among any pair of points.

II. RELATED WORK

A. Static CEUS Imaging Segmentation

Earlier work dedicated for CEUS imaging segmentation still rely on a specified reference frame in which lesions are sufficiently presented and well-distinguished from surrounding tissues. Generally, the reference frame is identified around the point of perfusion peaking. Bakas et al. [12] proposes a two-stage segmentation framework for CEUS imaging, which first localizes the focal liver lesion using active ellipse model and then refines the boundary by iteratively classifying the surrounding pixels. Furthermore, Meng et al. [13] proposes a dual-modality segmentation network CEUSegNet with the similar dual-branch top-down structure, which takes the pair of US and CEUS images as inputs and fuses the cross-modality features at each encoding scale. Overall, the segmentation performances of the two former methods largely depend on the selected reference frame whether a well-distinguished enhancement pattern is displayed. Due to the lack of US view, our model places the emphasis on temporal modeling of dynamic enhancement process rather than depending on the single enhancement pattern at certain point.

B. Dynamic CEUS Imaging Segmentation

In order to exploit the pattern of contrast enhancement over time, another line of work starts to treat the segmentation task as the classification of pixel- or patch-wise TICs. For example,

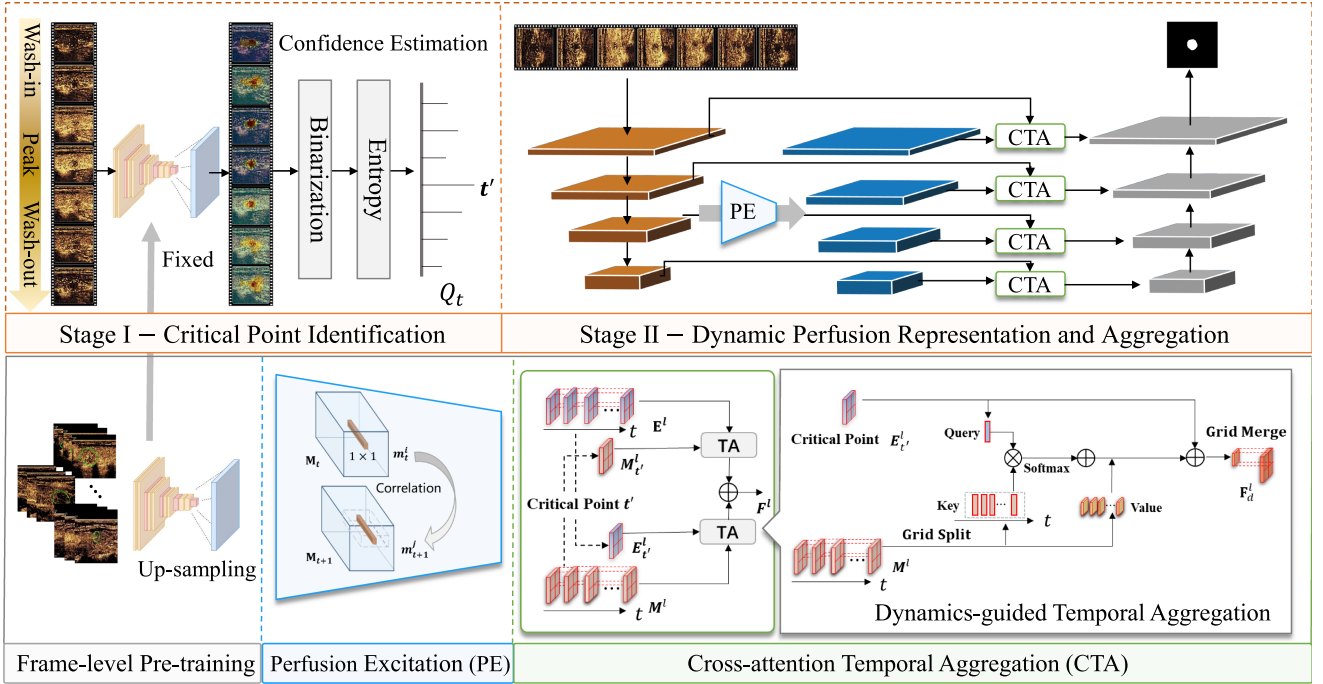


Fig. 2. A schematic diagram of the proposed two-stage dynamic perfusion representation and aggregation network (DpRAN). At the stage I, we adopt an entropy-based confidence estimation strategy to assist the identification of the critical point t' . One point is assumed to sufficiently present the boundaries of nodules in comparison to the rest frames. At the stage II, the enhancement morphology and dynamics features are encoded simultaneously (drawn in orange and blue, respectively). To capture the global enhancement dynamics, we design a cross-attention temporal aggregation (CTA) module which contextualizes enhancement patterns centering around the identified critical point t' . For simplicity, we only illustrate the dynamics-guided temporal aggregation and omit its counterpart, i.e., morphology-guided temporal aggregation.

Julie et al. [14] presents an interactive segmentation method which gathers pixel-wise perfusion curves into clusters via self-organizing map, and specifies the cluster type by expert intervention. Similar clustering-based pixel TICs grouping is adopted in another work of Bakas et al. [15]. Feng et al. [16] leverages 3D convolution neural network (CNN) to jointly model the spatial and temporal enhancement features of local patches for prostate cancer detection. Compared with existing static methods, these approaches move towards the analysis of dynamic enhancement characteristics of different tissues. Nonetheless, pixel- or patch-wise tissues classification inevitably ignores the neighboring perfusion information, which might lead to worse results in case of heterogenous enhancement. Based on that, we propose a novel dynamic perfusion representation and aggregation network which focuses on the frame-level enhancement pattern representation and dynamic enhancement evolution modeling for CEUS imaging segmentation over the whole CEUS sequence.

C. Temporal Information Modeling

A common temporal modeling strategy is to process each frame independently using 2D CNN, and aggregates temporal information with various mechanisms [17], [18], [19], [20]. To further exploit inter-frame correlations, the popular two-stream structure considers the static features of each point and motion information among adjacent frames simultaneously [21], [22], [23], [24]. It inspires us to explore blood flow information by explicitly modeling inter-frame differences. Besides, we find

that attention mechanism [25], [26], [27] is well suitable to capture the long-range enhancement evolution information by building associations among local points. In terms of video understanding, it is important to relate what happened at the current point to the associated scenes happened in the past and future [28], [29], [30].

III. METHOD

A. Overall Architecture

Fig. 2 illustrates the overall architecture of our proposed dynamic perfusion representation and aggregation network (DpRAN). It takes the sequence of dynamic enhancement patterns as inputs, and outputs a binary mask contouring target lesions. Summarily, we decompose the segmentation process into the two stages detailed as follows.

At the stage I, we adopt a confidence estimation strategy to locate the critical enhancement point with the relatively distinguished boundary. At the stage II, we model the enhancement morphology and dynamics features simultaneously, and aggregate them with the guidance of the identified critical enhancement point. In this way, we expect to distill dynamic enhancement discrepancy between lesion and healthy tissues sufficiently. It is noted that the specific design of DpRAN model is inspired by clinical practices where the so-called 'optimal' frame with the highest diagnostic value is first determined for the further perfusion analysis.

B. Critical Point Identification

To determine the critical point which maximizes the contrast between nodules and surrounding tissues, we first pre-train a frame-level segmentation network using Dice loss (described in Section III-F), which performs coarse segmentation at the first stage. In contrast to DpRAN model, we preserve the branch of enhancement morphology encoder and bilinearly up-sample the output to the full image resolution. Then we fix its weights to obtain the pixel-wise probability map $\mathbf{P}_t$ at point t . Based on that, the segmentation confidence score $\mathbf{Q}_t$ is calculated as the entropy of the predicted area:

$$\mathbf{Q}_t = \frac{1}{HW} \left(\sum_{h,w} \sum_c -\mathbf{P}_t(h, w, c) \ln \mathbf{P}_t(h, w, c) \right) \quad (1)$$

where $\mathbf{P}_t(h, w, c)$ is the probability value at location (h, w) for the class $c \in \{0, 1\}$. To calculate the confidence score, the threshold $\tau = 0.5$ is applied to binarize $\mathbf{P}_t$. From the definition of entropy, a lower $\mathbf{Q}_t$ means less uncertainty on the current prediction, that is, the segmentation confidence is higher. On the contrary, we could infer that there are lacking definite enhancement boundary or more heterogenous enhancement at that pint. Thus we locate the critical point by $t' = \arg \min \mathbf{Q}_t$.

C. Perfusion Excitation Gate

As for contrast-enhanced US, enhancement intensity variations correlate with the real-time hemodynamic changes within tumor and healthy tissues. The apparent changes of blood flow signal relate to the underlying factors, i.e., irregular branching, varying diameter, and high tortuosity, etc., which benefits to distinguish the tumor from normal tissues [31], [32]. Therefore, we introduce the perfusion excitation (PE) gate (as shown in Fig. 2) to encode the enhancement variations among adjacent frames.

Given the sequence of CEUS frames, enhancement morphology encoder learns the appearance features for each frame, producing the representation sequence $\mathbf{M} = [\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_T] \in \mathbb{R}^{T \times C \times H \times W}$.¹ For the adjacent time points t and $t + 1$, our PE gate models the short-range enhancement variation by comparing morphology feature similarity among neighboring positions. For example, given patch representation $\mathbf{m}_t^i$ at the position i , feature similarity is calculated with those patches of neighboring positions $j \in \Omega(i)$ at the next time step $t + 1$,

$$c_t(i, j) = \text{Conv}(\mathbf{m}_t^i, \mathbf{m}_{t+1}^j) \quad (2)$$

where Conv denotes the convolution operation, i, j are the 2-dimensional coordinates (i_x, i_y) . The maximal displacement of neighborhood $\Omega(i)$ is $k = 21$, i.e., $\Omega(i) = (i_x - 10, i_x + 10) \times (i_y - 10, i_y + 10)$. Feature similarity is calculated by the dot-product of local path representation implemented by convolving the patch representation $\mathbf{m}_t^i$ with $\mathbf{m}_{t+1}^j$. By enumerating all combinations of patch pairs, we could obtain a k^2 -dimensional correlation vector c_t^i for each

position i . Furtherly, we align the dimension of correlation vector to the channel of morphology representation $\mathbf{M}_t$ by linear projection. To ensure the spatial layout of dynamics representation $\mathbf{E}_t$ to align with that of $\mathbf{M}_t$, we both set the patch size and stride to 1. We also append an all-zero correlation map $\mathbf{E}(T) = \mathbf{0}$ for temporal consistency. It is assumed that a higher correlation indicates the less perfusion variations, and vice versa. More importantly, this manner allows to model enhancement dynamics features in accordance with the learning of morphology features.

D. Cross-Attention Temporal Aggregation Module

In clinical practice, radiologists usually initialize the nodule areas with the manually selected reference frame, and then observe the continue enhancement to outline the more fine-grained boundary. It inspires us to distill perfusion discrepancy between nodules and their adjacent parenchyma by relating enhancement characteristics centring around the identified critical point. Therefore, we propose a cross-attention temporal aggregation (CTA) module for long-range enhancement feature fusion. As shown in Fig. 2, this module takes the identified critical enhancement point as the guidance signal to fuse the dynamic information. We elaborate this module from the two aspects, i.e., regular grid split and cross-view fusion.

1) *Regular Grid Split*: The aggregation of enhancement information is achieved by a weighted summation after attending important temporal supports, which is similar to most video understanding work. Nonetheless, we found that assigning a single weight to every time point might be insufficient in the pixel-wise segmentation task. An intuitive example is that the onset of significant enhancement features of different tissues is likely to be inconsistent, i.e., intratumor heterogenous enhancement and sustained enhancement of surrounding tissues may appear in different points. In view of this, we choose to split morphology/variation maps $\mathbf{M}(\mathbf{E})$ into non-overlapped $r \times r$ regular grids, such that temporal weights are individually estimated for each grid (r is set to 4).

2) *Cross-View Fusion*: For each grid, we adopt the cross-view attention mechanism to attend important points. To enhance inter-view interaction, cross-view attention mechanism⁴ treats the critical enhancement morphology and dynamics representations $\mathbf{M}_{t'}(\mathbf{E}_{t'})$ as the guidance signal of the other view.

Morphology-guided temporal aggregation: Given the critical morphology map $\mathbf{M}_{t'}^l$ ² and the spatially aligned enhancement dynamics representation sequence $\mathbf{E}^l = [\mathbf{E}_1^l, \mathbf{E}_2^l, \dots, \mathbf{E}_T^l]$, we first squeeze the feature maps $\mathbf{M}_{t'}$ and $\mathbf{E}^l$ using global average pooling, $e_t^l = \text{GAP}(\mathbf{E}_t^l) \in \mathbb{R}^d$. To preserve the temporal ordering information, we then add d -dimensional temporally positional encoding pe into the squeezed dynamics representation [26].

$$\begin{aligned} pe(t, 2i) &= \sin\left(t/10000^{2i/d}\right) \\ pe(t, 2i + 1) &= \cos\left(t/10000^{2i/d}\right) \end{aligned} \quad (3)$$

¹For simplicity, we omit the superscript l here that denotes the down-sampling scale.

²For simplicity, we omit the superscript r that denotes the spatial grouping.

where t and i denote the temporal and dimensional index, respectively. After that, the corresponding query (Q) and key (K) feature in attention mechanism are generated using linear projection. Since the classification task is performed at the pixel level, we use 1×1 Conv to obtain the 2-dimensional value (V) map $\mathbf{E}_{v,t}$ to preserve the spatial layout of the learned enhancement representation map. We formulate the aggregation operation as follows:

$$\hat{\mathbf{E}} = \sum_{t=1}^T w_a(q_t, k_t) \mathbf{E}_{v,t} \quad (4)$$

$$w_a(q_t, k_t) = \frac{\exp(\langle q_t, k_t \rangle / \sqrt{d_{in}})}{\sum_{t=1}^T \exp(\langle q_t, k_t \rangle / \sqrt{d_{in}})} \quad (5)$$

where q_t and k_t are the query and key values with dimension of $d_{in} = 32$. Accordingly, morphology-guided temporal aggregation is calculated as the weighted summation of value maps, where weights are calculated by the dot-products of the query Q with all keys K, normalized by *Softmax* function.

Finally, we fuse the aggregated representation into the critical morphology map $\mathbf{M}_{v'}$ with the goal to contextualize enhancement patterns centering around the identified critical point. Similar idea could be referred to the video understanding work [29].

$$\mathbf{F}_a = \mathbf{M}_{v'} + \hat{\mathbf{E}} \quad (6)$$

Dynamics-guided temporal aggregation: In parallel, we have a natural choice for the query(Q), key(K) and value(V) tensors for enhancement morphology feature aggregation: dynamics representation at the critical point $\mathbf{E}_{v'}^l$ is the source of query, and the sequence of morphology map $\mathbf{M}^l = [\mathbf{M}_1^l, \mathbf{M}_2^l, \dots, \mathbf{M}_T^l]$ is the source of key and value sets. Similarly, we obtain the fused enhancement representation as follows,

$$\mathbf{F}_d = \mathbf{E}_{v'} + \sum_{t=1}^T w_d(q_t, k_t) \mathbf{M}_{v,t} \quad (7)$$

where $\mathbf{M}_{v,t}$ is the value map of enhancement morphology, w_d is the normalized temporal weight calculated as Eq.5 that indicates the contribution of each point to discriminate lesion from normal tissues.

E. Network Details

The enhancement morphology encoder is composed of an input convolution and four sequential residual convolution blocks, separated by 2×2 max-pooling layer. Each block consists of two 3×3 Conv layers (unit stride with zero-padding), followed by a batch normalization and ReLU activation layer. The number of channels is [32, 32; 32, 32; 64, 64; 128, 128]. Similarly, the decoder comprises four residual blocks separated by one deconvolution layer. The residual blocks merge and refine the information from the aggregated enhancement representation and the up-sampled feature map from the last scale. The number of channels is [128, 128; 64, 64; 32, 32; 32, 32]. Finally, we obtain the full-resolution category map $\mathbf{P}$ using 1-channel 1×1 Conv and Sigmoid activation layer.

TABLE I

DEMOGRAPHIC INFORMATION AND VIDEO PARAMETERS OF THE COLLECTED DATASET

Datasets	No.	Gender(F/M)	Age	Size(cm)	Duration(min)	Resolution(pixel)
Thyroid nodules	145	62/83	45.8	1.72	2.51	330×360

F. Implementation

The proposed networks were implemented using deep learning framework Pytorch 1.8.1, and run on a single GPU (NVIDIA TITAN RTX, 24 GB). Both for model training and pre-training, network parameters were updated using stochastic gradient descent with momentum, implemented by a default Adam optimizer. In our experiments, we adopt a dynamic learning rates (LR) adjustment. The initial LR is set to 0.0002 both for the pre-training and training stages. The minimal LR is set to the half of initial value, and the batch size is set to 1. The maximal number of pre-training and training epochs was set to 50 and 60, respectively, and the training process was terminated when validation performance reach the best. In case of the class-imbalance problem, our network was trained using the Dice loss,

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^N p_i y_i + \varepsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i + \varepsilon} \quad (8)$$

where N is the number of pixels in the image, $p_i \in [0, 1]$ is the predicted probability of i_{th} pixel belonging to the lesion area, $y_i \in \{0, 1\}$.

IV. MATERIALS

A. Dataset

In this study, we collected the CEUS datasets of thyroid nodules for model evaluation. The total 145 CEUS videos were collected on consecutive patients attending Nanjing Drum Tower Hospital from September 2017 to July 2020. Detailed information was listed in Table I. All thyroid examinations were performed on a Logiq E9 scanner (GE Healthcare, Milwaukee, WI, USA). The injected ultrasound contrast agents (UCAs) was the second-generation gas-filled microbubble contrast SonoVue (Bracco SpA, Milan, Italy) dissolved in 5 mL saline. To avoid microbubbles disruption, a low mechanical index ($MI \leq 0.12$) was set, entering the contrast mode. The frequency of ultrasound scanner is 3.5 MHz. Each examination was performed by an expert radiologist with over 5-year clinical manipulation experience, the scanning acoustic window and depth were adjusted simultaneously to display the longitudinal section of the tumor and its surrounding parenchyma. Dynamic CEUS imaging were stored in the ultrasound workstation and exported as AVI video files. Each video lasts around 2 minutes with the frame rate of 9 frames per second, recording the blood perfusion from baseline to the eventual wash out. In our experiments, each CEUS data was annotated by two senior radiologists to reduce inter-observer variabilities. Specifically, each radiologist reviewed the whole CEUS sequence back and forth, and then contoured the lesion boundary based on the whole perfusion examination. Approval

was obtained by the ethics review board of local hospital and the informed consent was obtained from patients before this study.

B. Preprocessing

Compared with the high temporal resolution of our collected CEUS videos, the changes of enhancement morphology tend to be slow and smoothing. We refer to the temporal pruning strategy proposed in [33], which calculates the inter-frame enhancement differences to extract the most informative frames. In this paper, we adopt the Gray-level co-occurrence matrix (GLCM) to estimate inter-frame difference Δf among adjacent frames and select the former 7 frames with the largest texture differences. Note that, we rearrange them in the temporal order.

V. EXPERIMENTS AND RESULTS

In this part, we first compare our DpRAN method with several state-of-the-art methods (Section V-C). Then, we test the effectiveness of two fundamental components in this method, including perfusion excitation gate (Section V-D) and cross-attention temporal aggregation module (Section V-E). After that, we further evaluate the influence of several important model hyper-parameters, including the number of spatial grouping (Section V-F). Finally, we provide the visualization results of model prediction and intermediate feature maps (Section V-G).

A. Experimental Setup

In this study, we split the collected CEUS videos into five equal-size folds, and select one fold as the testing set. Then, the remaining four folds are further split into training and validation sets (ratio: 3:1). By repeating experiments for five times, we obtain the mean score and standard deviation of all metrics. Quantitative segmentation metrics include Sensitivity (SEN), Specificity (SPE), Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and Hausdorff distance (HD) [34]. The former two metrics measure the degree of overlap between the prediction result S and ground truth Y , and the HD measures the boundary distances. The former two indexes are defined as:

$$\begin{aligned} \text{IoU} &= \frac{|S \cap Y|}{|S \cup Y|} \\ \text{DSC} &= \frac{2|S \cap Y|}{|S| + |Y|} \end{aligned} \quad (9)$$

where $\cap$ and $\cup$ denote the intersection and union sets of network predictions and ground truth, respectively. And HD is defined as:

$$\begin{aligned} \text{HD} &= \max \{d_{SY}, d_{YS}\} \\ &= \max \left\{ \max_{s \in S} \min_{y \in Y} d(s, y), \max_{y \in Y} \min_{s \in S} d(y, s) \right\} \end{aligned} \quad (10)$$

where $d(\cdot, \cdot)$ is the Euclidean distances between point sets. For DSC, the larger value indicates the better overlapping (within [0,1]), while the lower HD value indicates higher boundary similarity. For all experimental comparisons, we computed the p-value with the two-sample t-test.

B. Competing Methods

1) *Static Segmentation Methods*: The proposed DpRAN method was first compared with several classic segmentation methods, including 1) Fully convolutional network (FCN) [35]; 2) Unet [36]; 3) DeepLabv3+ [37]; 4) Cascade Unet [38]. Then, we also compare it with several state-of-the-art attention-based methods mostly for the global context modeling, including 1) Global context network (GCNet) [39]; 2) SEgmentation TRansformer (SETR) [40]; 3) SeMask [41]; and 4) SegFormer [42]. To fully exploit dynamic enhancement information, we selected three representative frames, i.e., the point of perfusion peak as well as two points (wash-in/wash-out stage) at which average perfusion intensity reaches 50% of the peak, convert them into gray-scaled images and concatenate them along channels as network inputs. In our experiments, FCN and GCNet were constructed on the classic encoder-decoder structure with dense residual convolution units. SETR adopted a progressive up-sampling strategy in [40] to reconstruct the 2D mask. As for SeMask and SegFormer, the input CEUS frames are divided into patches of size 4×4 using a non-overlapping and overlapping split manner, respectively. And then local patches are fed into hierarchical transformer encoder to generate features with the size of $\{\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}\}$ of original resolution. These hierarchical features are rescaled and concatenated for 2D mask reconstruction using a MLP-based decoder.

2) *Video Segmentation Methods*: Our DpRAN method was further compared with four video object segmentation (VOS) methods, including 1) recurrent network-based models (ConvLSTM and Bi-ConvLSTM [43]) that exploit temporal evolution with long short-term memory (LSTM) units [44]; 2) Motion guided attention (MGA) network [45], which attends and enhances morphology features with motion saliency map; 3) Co-attention Siamese Network (COSNet) [46] which adopts a Siamese architecture to capture cross-frame context correlations. It should be noted that general video object segmentation methods are designed to segment the object for every frame, in this paper, we adopt a late-fusion mechanism to merge probability maps of various frames via mean average. Also, we adjust the model parameters to control the model size.

C. Comparison With State-of-The-Art Methods

In this group of experiments, we compare the segmentation performance of our DpRAN with several state-of-the-art methods that cover the static and dynamic segmentation tasks. Quantitative results in terms of SEN, SPE, DSC, IoU and HD are presented in Table II. We have also visualized segmentation results of various methods in Fig. 5.

From this table, we could have a clear observation that video-based segmentation methods (i.e., Bi-ConvLSTM, COSNet, and our DpRAN) generally achieve evident performance improvements over those static approaches (i.e., FCN, Unet, DeepLabv3 and GCNet), even though CEUS frames from multiple perfusion phases are concatenated as inputs. As can be seen, the proposed DpRAN and COSNet rank the first and second place with IoU hitting 0.676 and 0.648, respectively. Among which, our method and COSNet achieve the highest specificity and sensitivity of

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TASK OF THYROID NODULE SEGMENTATION

Methods	Thyroid nodule				
	SEN	SPE	DSC	IoU	HD
FCN	0.761±0.021*	0.973±0.009*	0.736±0.016*	0.586±0.020*	7.415±0.587
Uet	0.773±0.026	0.972±0.007	0.741±0.014*	0.593±0.015*	7.419±0.342
Cascade Unet	0.764±0.011*	0.977±0.013	0.767±0.016*	0.626±0.013*	7.438±0.454
DeepLabv3	0.761±0.028*	0.973±0.014	0.745±0.015*	0.597±0.020*	7.805±0.475*
GCNet	0.748±0.033*	0.977±0.008	0.753±0.014*	0.607±0.017*	7.604±0.530*
SETR	0.772±0.022	0.974±0.010	0.760±0.016*	0.619±0.020*	7.506±0.494
SeMask	0.786±0.017*	0.969±0.013*	0.751±0.013*	0.606±0.016*	7.722±0.447
SegFormer	0.779±0.025	0.972±0.015	0.763±0.017*	0.621±0.015*	7.724±0.554
ConvLSTM	0.785±0.021	0.964±0.006*	0.757±0.015*	0.613±0.018*	8.128±0.517*
Bi-ConvLSTM	0.773±0.017	0.975±0.008	0.763±0.013*	0.621±0.017*	7.764±0.510
MGA	0.771±0.017	0.977±0.005	0.776±0.016*	0.639±0.019*	7.643±0.431
COSNet	0.796±0.035	0.958±0.014*	0.782±0.015	0.648±0.018	6.833±0.572*
Ours	0.778±0.013	0.979±0.003	0.794±0.027	0.676±0.015	7.120±0.460

* Denotes a significant difference (t-test, $p \leq 0.05$) compared with our method.

Boldface type represents $p > 0.1$ compared with the last row Ours.

0.979 and 0.796. The slight inferior performance is obtained by MGA with IoU of 0.639, which is still superior to most of static methods. Most of these improvements are statistically significant especially in terms of SEN, DSC and IoU. The optimal performance of static segmentation methods is obtained by Cascade Unet and SegFormer with DSC and IoU hitting around 0.76, and 0.62. The highest segmentation sensitivity of static methods is obtained by SeMask with 0.786 but its DSC and IoU scores are slightly inferior to them. As for baseline FCN, the IoU score drops to 0.586, which is much smaller than the rest methods. These results indicate that the concatenation of multi-phase CEUS frames along channels is less effective than aggregating dynamic enhancement features, and temporal modeling throughout the whole perfusion sequence is proved to be necessary for discriminating nodules and adjacent tissues. Also, we find that exploiting inter-frame collaborative effects with some attention mechanism (i.e., MGA, COSNet, and our DpRAN) outperform the conventional temporal accumulation methods with memory units (i.e., ConvLSTM and Bi-ConvLSTM). As for static methods, self-attention mechanism designed with various transformer (i.e., SETR, SeMask and SegFormer) structures also confirms its superiority over the most of conventional convolution networks (i.e., FCN, Unet, and DeepLabv3).

D. Comparison of Enhancement Dynamics Modeling

As introduced in Section III-C, the proposed perfusion excitation (PE) gate is specifically designed to exploit inter-frame perfusion variations for short-range enhancement modeling. Similar to motion features, enhancement features are expected to benefit object recognition. To validate its effectiveness, we compared it with two baselines and three other mainstream enhancement (motion) modeling methods. Baselines are actually for ablation analysis, where the enhancement encoder reduces to a single-stream structure. Also, the cross-view query

signal in temporal aggregation module degrades to the critical enhancement map of the input view.

- *Baseline1/2*: We remove or only preserve the enhancement dynamics encoder in the Baseline 1 and 2, respectively. Considering the reduced number of model parameters, we double the channels of the single-path encoder for performance comparison.
- *Optical Flow*: We implement the optical flow estimation as a forward flow from the previous frame to the current frame as in MGA [45], [47], and feed 2-channel optical flow images as the input of enhancement modeling branch, which shares the same structure as that of morphology learning.
- *Temporal Difference*: To model enhancement variations at the feature level, temporal difference method [48] simply derives the perfusion difference map via element-wise subtraction.
- *Motion Excitation*: To strengthen enhancement-sensitive channels, motion excitation method [11] recalibrates enhancement representations based on inter-frame feature difference maps.

From Table. III, we could have the following observations: 1) Fusing the feature maps from enhancement morphology and dynamics achieves significantly better results than two baselines. We find that Dice coefficient drops to 0.746 when only enhancement morphology was represented for Baseline 1, whereas Baseline 2 obtains the lowest Dice coefficient 0.733 when we only model enhancement variations among adjacent frames. These results verify the effectiveness of dual-view encoding of enhancement features both at the single point and successive points; 2) Correlating pair of adjacent frames at each spatial location (PE gate) outperforms the prevailing motion modeling methods by a large margin, either by pre-computed optical flow information or deep feature differences. We find that the manner of temporal difference achieves the highest mean sensitivity of 0.812 but its DSC and IoU are relatively lower with 0.757 and 0.618, respectively. Also for Motion Excitation, its

TABLE III
COMPARISON OF DIFFERENT METHODS OF ENHANCEMENT DYNAMICS MODELING ON THE TASK OF THYROID NODULE SEGMENTATION

Methods	Thyroid nodule				
	SEN	SPE	DSC	IoU	HD
Baseline1	0.790±0.024*	0.961±0.008*	0.746±0.012*	0.603±0.014*	8.203±0.544*
Baseline2	0.772±0.013	0.966±0.006*	0.733±0.015*	0.585±0.018*	8.462±0.476*
Optical Flow	0.779±0.013	0.976±0.004	0.769±0.010*	0.631±0.011	7.523±0.414
Temporal Difference	0.812±0.019*	0.955±0.007*	0.757±0.011*	0.618±0.014*	8.586±0.402*
Motion Excitation	0.787±0.018	0.970±0.009	0.749±0.016*	0.603±0.019*	7.808±0.605*
Ours	0.778±0.013	0.979±0.003	0.794±0.027	0.676±0.015	7.120±0.460

* Denotes a significant difference (t-test, $p \leq 0.05$) compared with our method.

Boldface type represents $p < 0.05$ compared with the last row Ours.

TABLE IV
COMPARISON OF DIFFERENT TEMPORAL AGGREGATION STRATEGIES ON THE TASK OF THYROID NODULE SEGMENTATION

Methods	Fusion	FLOPs (G)	Thyroid nodule				
			SEN	SPE	DSC	IoU	HD
TC	Early	21.328	0.792±0.038*	0.951±0.015*	0.738±0.017*	0.592±0.018*	8.788±0.519*
	Late	22.089	0.782±0.019	0.972±0.003	0.758±0.015*	0.615±0.021*	7.806±0.604*
	HyperDenseNet [50]	29.650	0.781±0.021	0.971±0.004	0.773±0.013	0.633±0.019*	7.712±0.423*
	Co-learning [51]	25.460	0.773±0.013	0.976±0.003	0.780±0.015	0.647±0.017	7.246±0.575
Non-local TA	Early	21.644	0.790±0.024*	0.967±0.013*	0.760±0.015*	0.616±0.019*	7.833±0.584*
	Late	20.868	0.773±0.016	0.976±0.009	0.779±0.014	0.642±0.018	7.607±0.388*
	HyperDenseNet [50]	29.646	0.772±0.022	0.971±0.011	0.768±0.013*	0.627±0.018*	6.904±0.388*
	Co-learning [51]	25.661	0.769±0.013*	0.975±0.010	0.787±0.015	0.657±0.015	7.143±0.493
Ours	CTA	21.783	0.778±0.013	0.979±0.003	0.794±0.027	0.676±0.015	7.120±0.460

* Denotes a significant difference (t-test, $p \leq 0.05$) compared with our method.

FLOPs — Floating point operations per second that measures computational complexity.

Boldface type represents $p < 0.05$ compared with the last row CTA

segmentation performance is much inferior to other dynamics modeling methods, reaching the DSC and IoU of 0.749 and 0.603, respectively. These results demonstrate the superiority of similarity comparison within a neighborhood of enhancement morphology representation; 3) In spite of a similar way to motion (variations) estimation, our method achieves a significantly higher Dice score of 0.794 and IoU of 0.676. It is perhaps due to that pre-computed flow information does not have the benefits of feature-level enhancement variations modeling, which is likely to be less coordinated with enhancement morphology representation learning.

E. Comparison of Temporal Aggregation Strategies

As introduced in Section III-D, cross-attention temporal aggregation (CTA) module that contextualizes temporal supports centering the identified critical point is the fundamental component for enhancement dynamics fusion. To verify its effectiveness, we compared it with another two types of temporal aggregation (TA) approaches.

- *Temporal Convolution (TC)*: Temporal aggregation is implemented by stacking a series of 1D convolution/pooling layers operated on the temporal dimension [18].
- *Non-local TA*: By contextualizing temporal supports at any pair of positions, Non-local TA models the long-range

temporal dependency at every point and strengthen similar semantics with each other [51].

In addition to temporal fusion, CTA module also achieves fusion of enhancement morphology and dynamics features via cross-view attention mechanism. Thus, we also adapt the temporal aggregation approaches by combining them with four types of view fusion mechanisms. These methods include 1) channel-wise concatenation before or after temporal aggregation, denoted as Early and Late fusion, respectively; 2) HyperDenseNet [49] that extends the dense connection to a multi-branch structure to facilitate information flow between modalities; 3) Co-learning network [52] that quantifies the relative importance of view-specific feature maps by deriving a co-learned fusion maps. Experimental results and the computational complexity of various variants as well as our CTA module are presented in Table IV.

From Table IV, we could observe that the proposed CTA module achieves competitive performance in the task of thyroid nodule segmentation. Most performance improvements brought by the designed cross-view fusion mechanism are statistically significant with respect to SEN, IoU, Dice coefficient, and HD. The highest sensitivity is obtained by the temporal convolution combined with early fusion with 0.792 but its DSC is much lower with 0.738, which is likely due to oversegmentation. Meanwhile, the results of the Non-local TA combined with

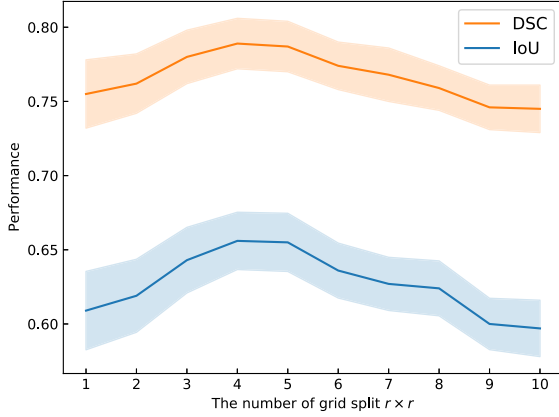


Fig. 3. Influence of the number of spatial grouping when tuning r from 1 to 10. Shaded region plots the standard deviation.

Co-learning fusion are slightly inferior to our method with SEN, DSC, and IoU of 0.769, 0.787, and 0.657. From the aspect of view fusion, Co-learning fusion shares the similar spatial weighting mechanism as that of our CTA module that involves spatial grouping. It confirms the superiority of group-wise temporal attention mechanism to capture enhancement dynamics of different regions. Nevertheless, no matter from temporal or spatial dimension, building pair-wise associations in Non-local TA or the pixel-wise weighting estimation in Co-learning is less computationally efficient compared with our CTA module.

Besides, fusing different enhancement features (morphology and dynamics) at the early level is the weakest manner to fuse cross-view complementarity for enhancement discrepancy extraction. The late fusion manner that merges temporally-aggregated features from independent paths achieved slightly superior performance in terms of IoU and Dice coefficient. In contrast, layer-level fusion by HyperDensenNet and Co-learning provides a clear improvement, increasing the IoU from 0.59(0.61) to 0.64(0.65). This shows that the manner of layer-level fusion could be the direct option for dual-view enhancement feature fusion along with temporal information aggregation.

F. Influence of the Number of Spatial Grouping

In this group of experiments, we investigated the influence of the number of spatial grouping in CTA module on the segmentation performance (denoted as $r \times r$). Using thyroid nodule segmentation as an example, we orderly selected r from [1–10] in CTA module and plotted the changing tendency in terms of DSC and IoU in Fig. 3.

Fig. 3 shows the changing tendency of metrics when changing the r from 1 to 10 for regular grid split. When changing r from 1 to 4 (the number of grids from 1 to 16), both the values of DSC and IoU are clearly increased within this interval, increasing IoU from 0.609 to 0.676, and DSC from 0.755 to 0.794. Then the segmentation performance begins to decrease continuously when r exceeds 5, with a performance degradation nearly 8%. It is naturally to interpret the initial steady increase ($r = [1 - 5]$) since the relative importance of enhancement features are quantified by a finer level, thereby taking the spatial-temporal discrepancy into

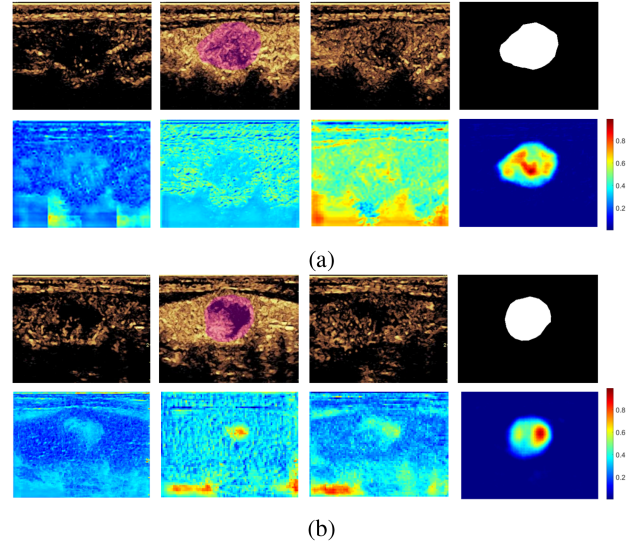


Fig. 4. Visualization results of our DpRAN method. The first row is the three representative perfusion points which cover the initial enhancement, reaching peak, and the eventual wash out, as well as the ground truth. Segmentation results are overlaid on the peaking frame. The second row is the intermediate fusion features generated by the last CTA module and the restored map before the final convolution layer. The first two columns are the morphology-guided temporal aggregation feature F_a^1 and dynamics-guided temporal aggregation feature F_d^1 corresponding to M^1/E^1 . And the third column are the combined features $F_a^1 + F_d^1$. For illustration, we squeezed features along channels and rescaled feature values into [0 – 1]. As observed, the intermediate fusion features in (a) and (b) illustrate clearly different feature responses between lesion and surrounding tissues. From the 2nd row 4th column of both subfigures, we could further observe the perfusion heterogeneity among lesion interiors.

account as possible. However, a denser spatial partition could not lead to further performance improvements, especially when the number of grids is over 50. The main reason could be that excessive grid split results in too small grids, which is unfit for part of perfusion regions which have the same homogenous perfusion patterns, meanwhile, the significantly increased computational complexity also affects model generalization. The reason of choosing $r = 4$ in current implementation is to sufficiently consider the importance difference of spatial-temporal units as well as to reduce the difficulty of network training.

G. Visualization Result

Towards a more intuitive understanding of our DpRAN method in the task of contrast-enhanced US based lesion segmentation, we provide a visualization of the intermediate morphology and dynamics feature fusion and the final model prediction in Fig. 4, which take the two thyroid nodules as examples. For each case, we choose to illustrate three typical enhancement frames that cover the initial enhancement, reaching peak and the wash out. Segmentation results are overlaid on the peaking frame. For the second row, the former two columns are colored maps of the morphology-guided temporal aggregation feature F_a^1 and dynamics-guided temporal aggregation feature F_d^1 , respectively, which are derived from the morphology/dynamics features M^1/E^1 after the first residual block. Column

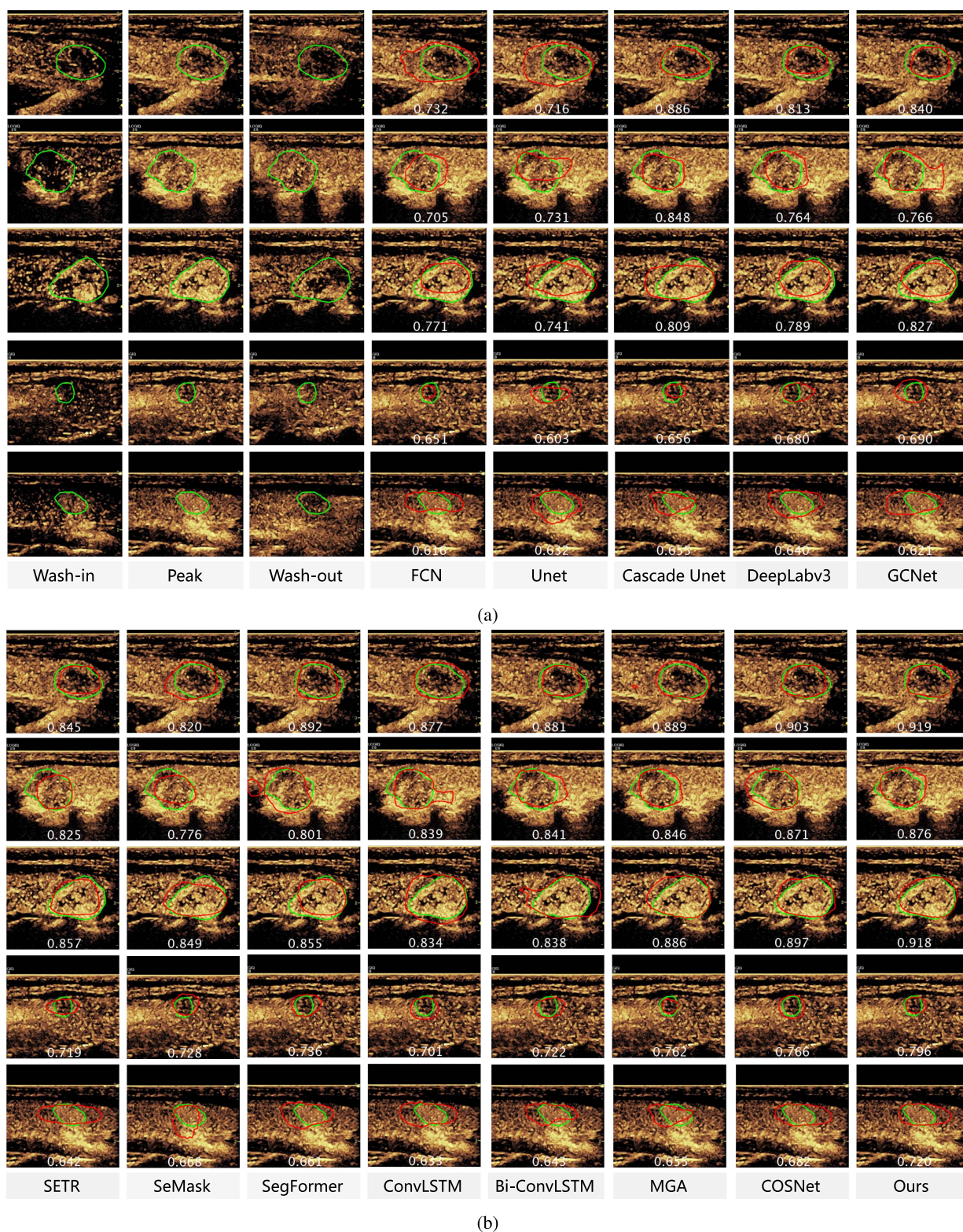


Fig. 5. Visualization of segmentation results of various competing methods. The first three columns are the three representative perfusion points which cover the initial wash-in, reaching peak, and the eventual wash out. The rest columns are the segmentation predictions overlaid on the peaking frames. Ground truths of thyroid nodules are contoured by green curves and the model prediction are contoured by the red. We also annotated the Dice coefficient of each prediction.

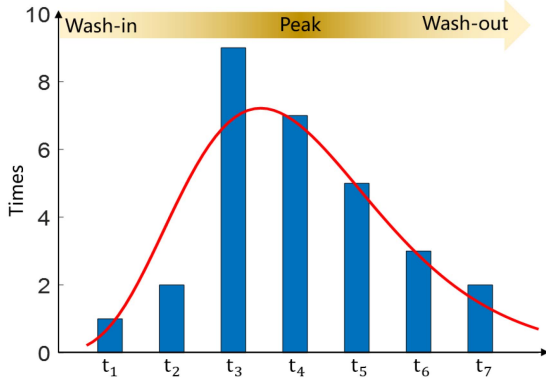


Fig. 6. Frequency count of the identified critical point among the input CEUS sequence. Red curve plots the probability fitting results of the statistics.

3 correspond to the visualization of $\mathbf{F}_a^1 + \mathbf{F}_d^1$. The last column is the reconstructed map before the final convolution. In this figure, each feature map has been squeezed along channels and independently normalized to $[0 - 1]$ for visualization. The figure shows the differences of feature responses of different perfusion regions, which combines the temporally aggregated morphology and dynamics features. From Fig. 4, we could see that, either for dynamics-guided temporal aggregation feature or morphology-guided temporal aggregation, feature responses around lesion regions are clearly different from surrounding tissues, especially for the lesion interiors. This validates that the varying evolution patterns are captured by our CTA module. From Fig. 4(b), we observe that, when there exists a defect filling centering thyroid nodule, distinguished feature responses are visible both in morphology and dynamics aggregation feature map, especially in the final category map (Column 4), since the perfusion discrepancies are much more evident in relative to healthy tissues with full enhancement.

In Fig. 5, we illustrate the segmentation results of various competing methods. The top three rows show the results of nodules with relatively large size and the bottom two rows correspond to results of much smaller nodules. We could find that, for different sizes of nodules, our method achieves consistent improvement over those competing methods. Overall, still segmentation methods show inferior segmentation accuracy than those temporal fusion-based methods.

H. Statistics of Critical Point

As introduced in Section III-A, the identification of critical enhancement point is the first step for the subsequent cross-view temporal aggregation. In Fig. 6, we plot the frequency count of the identified critical point for the test set ($n = 29$). We could clearly observe that most of the critical points distribute over the intermediate points, from t_3 to t_5 , nearly around perfusion peak. It is in line with our observation that typical perfusion patterns, such as boundary rim-like enhancement or distinguished partial defect, frequently occurs around peaking points. It also validates the effectiveness of frame-level pre-training for the identification of critical point t' .

VI. DISCUSSION

A. Limitations and Future Work

Although the superior performance of the DpRAN method has been validated on extensive experiments, there are still several limitations to be addressed in the current implementation. In summary, the core part is how to model and fuse dynamic enhancement characteristics efficiently. In our CTA module, we introduce the concept of spatial grouping in comparison with common cross-attention mechanism, with the goal to differentiate temporal weights across distinct regions. Nonetheless, regular spatial partition without overlapping still lacks enough flexibility to capture salient perfusion differences, especially for heterogeneous enhancement sequences. Thus, it is a promising direction to design a more flexible spatial grouping with varying shapes and sizes, which further improves the efficiency of temporal aggregation.

VII. CONCLUSION

In this work, we propose a two-stream U-shape segmentation network, DpRAN, for dynamic perfusion representation and aggregation for the task of CEUS-based lesion segmentation. The main innovation consists in the perfusion excitation gate designed for dynamics representation, and cross-attention temporal aggregation module for cross-view temporal fusion. On the collected datasets of thyroid nodule, our proposed DpRAN method achieved superior or at least comparable segmentation performance in contrast to state-of-the-art methods and various ablation analysis. It validates the effectiveness of the overall network structure for enhancement modeling in the task of dynamic CEUS-based segmentation.

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